

# Traffic Accidents Analysis Report Using Power BI

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**Abstract**—The Objective of this project is to give an analysis of the traffic accident dataset and evaluate the patterns and trends in it using Power BI Dashboard. Finding the key elements in the crash variables in the dataset, quantifying the impact of accidents, and providing executives with a simple environment for decision-making are the initiatives that are depicted in the dataset. Data cleaning, Intelligent Dax measures, time trend-based analysis, monetary effect evaluations, and decomposition tree-based root cause analysis are the steps in the process. Toggle buttons, tooltips, and dynamic measures that replicate every dashboard feature are all included in the report. For providing the better understanding of the information about the accident patterns, Severities, Damage Cost and Other Contributing factors, a professional report was developed.

## I. INTRODUCTION

Every year, a large number of people are getting killed or injured in traffic accidents, which has been a major public concern on a global scale. According to the World Health Organization (WHO), tens of millions of people suffer non-fatal injuries in motor vehicle accidents each year, while 1.19 million people are killed in these incidents [1]. These cause immense personal suffering for individuals and will be having impacts social and economic impacts on a normal person as they lead to lost productivity and huge chunks of medical expenses. Thus, using evidence-based information, transportation planners and policymakers have focused their efforts on improving road safety [2]. In order to mitigate the accident a large scale analysis is required on the traffic accident, which could lead to a potential solution. In the analysis part, identify the patterns causing the accidents, risk factors, and very risky areas so the policy makers can create a efficient safety measures. Road deaths can be reduced by using targeted mitigation measures made easier by identifying the causes of crashes.

Although a clear analysis of traffic accidents is obviously necessary, conventional methods of accident data reporting clearly create great challenges. Mostly massive and spanning many years and territory, accident databases gathered by police and transportation companies are Such data sets—which contain information about crash conditions, vehicles, weather, road surface, etc.—have size and large dimensionality that make hand analysis impossible. Furthermore common quality problems with data records might include human mistakes, missing values, or incorrect coding. From the previous studies, incomplete or inconsistent data can show wrong trends and provide a false findings[3]. For example, If you neglect the

details on weather or illumination can conceal their real impact on crash severity. Likewise, variations in reporting forms across areas prevent any comprehensive research from being possible. Expressing the complex interdependence of the variables like driver behavior, environment, and infrastructure, is difficult using conventional static reporting and spreadsheet-based analysis. Key information is buried in raw accident records, generally without good data preprocessing and analysis techniques. These issues highlight the need of contemporary tools and methods for securely cleaning, merging, and evaluating traffic accident statistics. In the same regard, deviations in reporting forms across regions render them unaffected for an exhaustive study. There are also certain details in surface-based analysis and static traditional reports that deal with complex interdependence of the variables such as driver behavior, environmental factors, and infrastructure. Generally, without proper preprocessing and analysis methods, important information is kept in raw records of accidents. These concerns accentuate the contemporary tools and techniques worth adopting in cleaning, integrating, and analyzing traffic accident data.

In this regard, Business Intelligence (BI) software has turned to the aiding solutions that convert raw traffic data into insightful information. BI technologies provide for an integrated environment for cleaning data, modeling, analysis, and interactive visualization. Rather than spend time in hand reporting, government agencies can use a series of BI dashboards to monitor accident trends and interesting performance metrics in the now. Recent research finds that the application of data science and business intelligence concepts to traffic safety data enables traffic authorities to perceive crash information within simple-to-generate visual representations and to make timely decisions [2]. The decision-makers can expose time trends (e.g., hours of day with most accidents or seasonal trends) and locations of spatial hotspot, and thus "drill down" to further examine specific instances, all supported by the capability to collect and filter the data in near real-time. This approach to dynamic descriptive analytics, rather than a static yearly aggregation, can query the data in real time. BI retrospective analysis lends itself to fast insights and evidence-based policy-making; thus is one way of open data saving lives and money. Converting of large volume accident data into a clear beautiful visual narrative is one big step in road safety management.

Microsoft Power BI is one of the new BI technologies in use today, and it is the most versatile for analyzing and creating

dashboards for traffic accident analysis. Extensive functionality, including data import and cleaning through Power Query, interactive visualization development, all fit within it. The data sources are numerous, as well as importing large datasets and transforming them with high-level transformations such as missing data filling and formatting. When the data model is created, analysts can generate additional measures that expand beyond raw data, such as accident severity rates and per car injury, with Data Analysis Expressions (DAX). This large visualization layer includes the slicers, filters, and drill-downs added in Power BI, which allows dynamic drill-down investigation of crash patterns over time, location, severity, and conditions. Such exploratory questioning reveals latent relationships that are not always manifesting in traditional reports. Using Power BI, uncertainty surrounding road accidents is transformed into actionable insights through its combined features of data cleansing, analytical modeling, and rich visualizations, and can benefit road safety experts and policymakers by facilitating strategic decisions [4].

## II. BACKGROUND AND RELATED WORK

Road traffic injuries are a persistent global issue, resulting in major deaths, injuries, and economic losses. Historically, accident data collection was manual through reporting by police departments and transportation authorities, which tended to result in incomplete or inconsistent data. Traditional analytical tools, such as simple tabular summaries and static reports, struggled to uncover useful information due to constraints in dealing with large datasets, missing values, and the multidimensional nature of crash events [1]. Early studies were concerned with simple frequency counts or correlation analysis, resulting in a narrow picture of the interconnected relationships between environmental variables, driver behavior, and road conditions. The introduction of data-driven methods and advanced computational software has transformed traffic accident investigation. Academics are now using machine learning, spatial analytics, and business intelligence (BI) platforms to identify hidden patterns in accident data sets[5]. Visualization-driven analytics have gained popularity, allowing stakeholders to easily examine trends such as collision severity by light situation, or accident frequency throughout weekdays and weekends[3]. Dynamic dashboards, in particular, provide a flexible environment for filtering, slicing, and drilling down into datasets without the need for technical skills, hence increasing access to data-driven decision-making. Because of its strong data cleansing, modeling, and visualization capabilities, Microsoft Power BI has emerged as a premier solution for large-scale accident investigation in the business intelligence arena. Power BI's Power Query enables users to preprocess and normalize complicated datasets, including missing values and datatype discrepancies. DAX (Data Analysis Expressions) allows for the creation of tailored measurements like crash rates, injury ratios, and time-based growth indicators. Recent research exemplifies the usefulness of BI dashboards for real-time monitoring of traffic events, high-risk spot identification, and impacting traffic safety policies [4]. Interactive Power

BI tools, including slicers and decomposition trees, allow stakeholders to dynamically analyze various causes of crash patterns, raise awareness, and guide focused intervention strategies. The application of BI technologies is expanding increasingly in transport analytics, challenges persist. Heterogeneity of data, privacy concerns, and real-time integration of data are persistent issues[6]. Moreover, although static dashboards offer crucial descriptive insights, integration of predictive analytics and AI-based modeling remains a persistent field of research [7]. Subsequent research will persist to emphasize the convergence of descriptive- dashboards and predictive models so as to visualize as well as predict traffic problems under dynamic scenarios. Power BI is leveraged in this present project not only for visualization capabilities, and for end-to-end data preparation, time intelligence modeling, finance effect analysis, and dynamic exploration of crash contributing factors. This method extends existing techniques while improving dashboard interactivity, depth of analysis, and visual narrative capability to inform road safety research and policy making more effectively.

## III. SYSTEM ARCHITECTURE

### A. Hardware and Software Setup

The traffic accident dashboard was developed and tested on a typical computing environment. The hardware platform was a laptop with an Intel Core i7 processor, 16 GB of system RAM, and a 512 GB solid-state drive, which provided adequate memory and processing capacity to handle reasonably large datasets without significant delay. Power BI can be scaled to enterprise-class servers and cloud platforms but, for this project's purpose, Power BI Desktop (version February 2024) was used for desktop development and testing. The data set used with over 200,000 records of crashes could be handled by the memory capacities of the desktop version. Reporting generation and processing operations did not require any additional high-performance hardware acceleration, such as GPU computing.

### B. Overall Pipeline

The overall process of creating the dashboard was systematic and modular. To begin with, raw traffic crash data were loaded into Power BI via Power Query. Data cleaning processes included type correction, removal of inconsistencies, normalization of category variables, and management of missing values. Subsequent to data cleaning, calculated columns were added to enrich the dataset, mostly the "Injury Severity Level," "Crash Weekday Name," and "Daylight Cycle" categories. After developing the data model, quite some DAX metrics were programmed to calculate accident frequencies, injury decompositions, damage estimates, and time-wise aggregations like year-to-date (YTD) and month-to-date (MTD) crash counts. Visualizations were then built upon this model, and this resulted in interactive report pages like an Executive Summary, Time Trend Analysis, Damage and Units Report, Decomposition Tree Analysis, and a Tooltip Page.

### C. Expanded Pipeline Stages and Data Flow

The architecture of data flow can be divided into various larger phases. The data ingestion phase first connects to the static accident CSV file via Power Query and performs transformations and cleansing activities. On ingestion, the data enrichment phase builds supplementary fields to enable better slicing, filtering, and aggregation. Post data preparation, the measure development step leverages DAX to build dynamic measures like crash severity distributions, damage trends over time, and unit-involvement rates. These measures are then integrated into the visualization stage, which includes interactive chart, slicer, and card design. The dynamic feature integration stage was treated with special attention, with toggle buttons and conditional tooltips being customized to facilitate user discovery. The final stage, dashboard publication, entails gathering all visuals and interactions into a single report ready for stakeholder examination or wider dissemination. Throughout the process, a modularity focus ensured that any alteration to the dataset would be achievable with minimum disruption to the dashboard layout

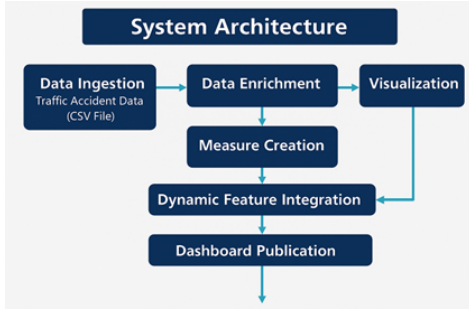


Fig. 1. Overall system architecture and data flow for the traffic accident dashboard.

## IV. METHODOLOGY

### A. Data Cleaning

The first step of the research was to ready the raw traffic accident data for analysis. Power Query Editor in Power BI was utilized to promote headers, fix data types, and replace invalid or missing data with standard statements like "Not Reported." Weather, road, and light conditions were all given specific attention because inconsistencies could distort trend analysis. Additional calculated columns were added as well to expand the analytical scope of the dataset. These included "Injury Severity Level," which coded crashes by the outcome of the injury; "Crash Weekday Name," which identified the day of the week; and "Daylight Cycle," which categorized the day light condition when the crash occurred. A custom Date Table was also constructed to facilitate intricate temporal intelligence calculations with the capacity to aggregate uninterruptedly by years, months, weeks, and quarters. Worldwide, the data cleansing process embarked to make a clean, coherent, and analytically ready dataset facilitating more trustworthy and informative reporting.

### B. DAX Measure Development

Following data preparation, a robust set of DAX (Data Analysis Expressions) measures was created to calculate key metrics to be visualized and analyzed. Measures included simple aggregations like total numbers of crashes and total injuries, and more complex calculations like proportion of injury crashes, average injuries per crash, and estimated total damage costs. Time intelligence functions like Year-to-Date (YTD), Month-to-Date (MTD), and Previous Year (PY) crash counts were used to analyze temporal trends. In addition, dynamic calculations were designed to allow users to switch between displaying total damage and total units engaged in crashes. Growth analysis calculations, such as year-over-year (YOY) and month-over-month (MOM) crash trends, provided insight into the way accident events evolved over time. These DAX measurements provided the analytical basis of the report, allowing rich and flexible data analysis.

### C. Dashboard Design and Interaction

The process ended with designing and creating a multi-page interactive dashboard. The report was organized into five sections: the Executive Summary, Time Trend Analysis, Damage and Units Report, Decomposition Tree Analysis, and Custom Tooltip Page. Visualizations included clustered column charts, area charts, pie charts, treemaps, decomposition trees, and dynamic KPI cards. Slicers and filters were strategically placed on each page to allow users to interactively explore the dataset using attributes like year, weather, road surface condition, lighting, and injury severity. The Damage and Units page now includes a toggle feature that allows you to dynamically switch between damage numbers and unit quantities. Custom tooltips with micro-trend sparklines were employed to show additional information on hover over visuals. The design prioritized clarity, interactivity, and seamless navigation, resulting in a professional and user-centered reporting experience.

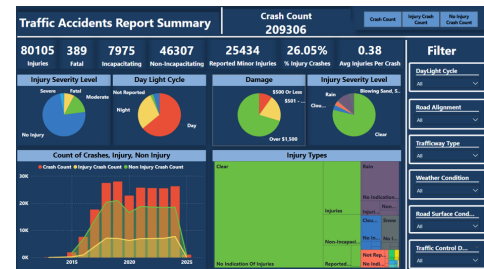


Fig. 2. Sample Dashboard of Executive Summary of the Data

## V. EVALUATION AND RESULTS

### A. Trends Observed

The research picked up on relevant trends in the traffic accident dataset. Temporal analysis showed that crash frequency rose in the fourth quarter of the year, particularly during October, November, and December. Further study by weekday revealed accidents to happen considerably more

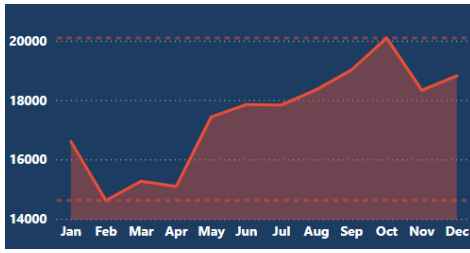


Fig. 3. Damages Crash by Month Plot

often on weekdays, with Friday being the most accident-prone day. Analyzing the collision severity split showed that the majority of occurrences bore no reported injuries, but a considerable number still suffered minor, non-incapacitating, and incapacitating injuries. Weather conditions and lighting conditions also played critical roles; good weather and daylight hours were linked with the highest number of crashes, while bad weather conditions such as rain and night illumination were responsible for a higher percentage of injury crashes rather than total crashes.

### B. Financial Analysis

The fiscal impact study, based on estimated damages costs recorded in the database, revealed that high-damage crashes constituted the vast majority of financial losses. Dynamic measurements allowed crash counts to be separated into three damage ranges: low (less than \$500), medium (\$501- \$1,500), and large. High-damage crashes accounted for the highest proportion, both in terms of number and monetary cost. Users could toggle between displaying total estimated damages and total units involved in collisions, revealing that, while units per crash stayed roughly constant, financial cost fluctuated greatly depending on crash severity and environmental conditions. Average damage per unit and average damage per crash were also calculated, for an even more detailed description of the economic impact of various collision types and conditions.

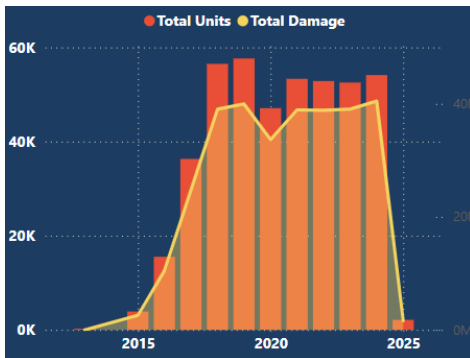


Fig. 4. Damages and units by year

### C. Decomposition Insights

Decomposition tree analysis revealed additional regarding the influences on crash frequency. Starting with total crash

numbers, the decomposition indicated that weather conditions, level and straight road alignment, and dry road conditions were the strongest influences on crash events. Further analysis indicated that crashes happened mostly in the absence of road faults and predominantly on weekdays during daylight hours. This multi-layered investigation revealed the interaction of numerous variables, providing crucial context beyond that which univariate analysis was able to offer. The deconstruction technique permitted stakeholders to dynamically examine "what-if" scenarios, i.e., how crash numbers would differ in rainy or icy weather compared to dry, clear weather. These findings underscore the need for taking environmental and highway conditions into account while planning for traffic safety and resource allocation.

## VI. LIMITATIONS AND CHALLENGES

Although the dashboard offers detailed information on traffic accident patterns, several limitations must be overcome. One major concern was the lack of consistency in recording estimated damage costs, potentially biasing financial analysis metrics. Damage assessments were missing or ambiguously categorized in many records and needed to be imputed or excluded from calculated metrics with defined values. Injury severity classification was also determined based on information available, but variations in reporting between cases could have resulted in slight misclassifications. Environmental variables such as lighting and weather conditions were frequently labeled as "Unknown," which may compromise the validity of conditional trend analysis. Another limitation is the dataset's static nature; it can represent only historical incidents and not dynamic factors such as traffic pattern modifications, road condition modifications, or policy implementations after data acquisition. Finally, while decomposition tree analysis provides useful exploratory insights, it is a descriptive technique that does not imply causal associations. Future improvements can include real-time data sets and predictive modeling techniques to transcend these limitations and yield even more valuable insight.

## VII. APPLICATIONS AND FUTURE WORK

The traffic accident dashboard built for this initiative has various applications in real-life transportation safety, urban planning, and insurance. Public safety officers can use the dashboard to track crash trends, identify peak-risk times and areas, and allocate road safety campaigns. Urban planners can use information into collision factors, such as road surface irregularities and illumination, to inform infrastructure improvement and traffic management techniques. Insurance organizations could benefit by examining crash seriousness trends and weather correlations for optimizing risk models and premium structure. There is immense scope in the future to increase the capability of the project.

Having real-time traffic crash feeds, facilitating dynamic updates of dashboards and live monitoring of incidents, would be a giant stride forward. Additional improvements would be the application of predictive analytics models, e.g., machine

learning crash severity predictors, that would shift the system from descriptive to prescriptive analytics. The integration of spatial analysis with GIS technologies could result in location-based crash heatmaps. Moreover, hosting the dashboard on a cloud platform such as Microsoft Power BI Service would introduce more collaboration, scheduled data refresh, and real-time alerting features, making the system more effective and easier to use for stakeholders.

## VIII. CONCLUSION

This project successfully demonstrated the creation of an end-to-end, dynamic traffic accident analysis dashboard through Microsoft Power BI. An easy-to-use, multi-aspect reporting system was developed through systematic data cleaning, enrichment, DAX measure development, and dashboard structure. The dashboard facilitates examination of crash quantities, injury severity distribution, estimated financial damages, and contributory environment causes along many facets and time periods.

Key outcomes, such as elevated rates of crashes during specific months and weekdays, the distribution of financial impact across collision severity classes, and essential environmental variables determined by decomposition analysis, highlight the utility of visual analytics in traffic safety research. Despite some limitations in data quality and static reporting that were noted, the project provides a solid base for continued progress toward real-time data, predictive modeling, and cloud deployment. Collectively, the article shows how business intelligence solutions can make raw facts about accidents powerful enough to inform in-depth insights towards creating smarter, safer communities.

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